

Orchestration Technique of Intelligent Microservices on the Edge Computing in High-Resource-Constraint Devices

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UFCG

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Outline

Orchestration
Technique of
Intelligent
Microservices
on the Edge
Computing
in
High-
Resource-
Constraint
Devices

Caio José
Borba Vilar
Guimarães

Objectives

Main Objective
Specific Objectives

State of the
Art

Methodology

Expected
Contributions

Project
Timeline

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 - Main Objective
 - Specific Objectives

- 2 State of the Art

- 3 Methodology

- 4 Expected Contributions

- 5 Project Timeline

Main Objective

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To develop, evaluate, and validate a technique for orchestrating intelligent microservices within a distributed and collaborative Edge Computing (EC) infrastructure composed of resource-constrained devices.

- To design a clear topology architecture for the distributed and collaborative Edge Computing infrastructure;
- To implement a collaborative and distributed scheduling system that intelligently instantiate microservices on connected devices;
- To create methods and algorithms for intelligent microservices orchestration;
- To test and evaluate these solutions both in simulated environments and real IoT setups, covering scenarios typical of Smart Environments and IoT applications.

Specific Objectives

- Study paradigms of orchestration techniques and protocols for collaborative distribution among constrained devices;
- Design and develop a technique tailored for high-restriction devices in distributed and cooperative environments, particularly in smart environments;
- Simulate existing state-of-the-art techniques in smart environments to evaluate their performance in terms of transmission rate, energy consumption, and scalability in microservices deployment;
- Develop an infrastructure with high-restriction devices and diverse communication technologies (wired and wireless) to test selected techniques in scenarios closely resembling real-world IoT environments;

Specific Objectives

- Implement and evaluate AI microservices (Machine Learning and Deep Learning models)—assessing accuracy, inference latency, and provisioning time in both simulated and real IoT environments;
- Develop new methods and algorithms for orchestration and cooperative control of intelligent microservices, optimizing the use of limited resources and mitigating the limitations of existing techniques;
- Analyze and compare the performance of simulation and real-world implementations, refining techniques and protocols based on empirical data;
- Ensure the developed protocols are feasible and conformant, applying formal analysis methods to verify their viability considering system constraints like memory and processing power;

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- Optimize existing algorithms and protocols by improving efficiency and reducing resource consumption, adapting based on feedback from initial implementations and tests

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- Edge Computing in Smart Environments: Emphasis on processing data locally at the network edge to reduce latency, bandwidth usage, and dependence on cloud services, enabling real-time applications in smart cities, industry, and surveillance;
- Challenges in Edge-Oriented IoT/IIoT Systems: Addressing issues like standardization, security, limited processing power, and model complexity of AI algorithms on low-resource devices;
- Distributed Orchestration of Microservices: Developing techniques for managing and coordinating microservices among resource-constrained devices without centralized control, improving scalability, robustness, and efficiency;

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- Collaborative Processing and Resource Sharing: Enabling multiple devices to work together collaboratively, composing services dynamically while considering constraints like latency, energy consumption, and network variability;
- AI and Machine Learning in Edge Devices: Implementing and evaluating intelligent models (ML and Deep Learning) directly on edge devices for real-time decision making, considering resource limitations and the need for local data processing;

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- Simulation and Validation in Realistic Environments:
Using simulation tools to assess techniques' performance before deployment in real IoT setups, ensuring viability and scalability;
- Protocols and Algorithms for Resource-Constrained Networks: Designing communication protocols and scheduling algorithms tailored to highly restricted devices operating cooperatively in dynamic environments

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- Literature Review of Existing Techniques;
- Design of Collaborative Edge Microservice Architecture;
- Development of Dynamic Collaborative Orchestration and Scheduling Algorithms;
- Implementation of Protocols for Device Communication;
- Simulation of Proposed Techniques in Controlled Environments;
- Performance Evaluation Using Metrics (latency, energy consumption, throughput);
- Validation in IoT and Real-World Environments;
- Comparative Analysis with Baseline Methods;
- Iterative Optimization Based on Test Results;
- Documentation.

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- Development of an innovative microservice orchestration technique for resource-constrained devices;
- Design of a scalable, collaborative edge computing architecture;
- Protocols for device communication agnostic of specific technologies;
- Algorithms for dynamic task scheduling and resource allocation;
- Performance benchmarks in simulated and real IoT environments;

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- Enhanced efficiency and reduced energy consumption in edge systems;
- Validation of microservice-based AI applications in smart environments;
- Contribution to scalable, real-time processing solutions for smart city applications;
- Framework for integrating high-restriction devices into distributed EC systems;
- Facilitation of interoperability among heterogeneous communication technologies.

■ 1st Semester

- Study of the state of the art (January 2024 – June 2024)

■ 2nd Semester

- Simulation of techniques (July 2024 – December 2024)

■ End of Year 1

- Research communication: present initial results and challenges (February 2025 – March 2025)

■ 3rd Semester

- Development of AI microservices (April 2025 – September 2025)

■ 4th Semester

- Testing in IoT environment (October 2025 – March 2026)

■ End of Year 2

- Research communication: present intermediate results and impact of optimizations (May 2026 – June 2026)

■ 5th Semester

- Qualification exam (July 2026 – December 2026)

■ 6th Semester

- Algorithm optimization (January 2027 – June 2027)

■ End of Year 3

- Research communication: present intermediate results and impact of optimizations (July 2027 – August 2027)

■ 7th Semester

- Protocol development (September 2027 – February 2028)

■ 8th Semester

- Thesis writing and revision (March 2028 – August 2028)

■ Thesis defense

- Defense of the thesis (September 2028)